

Same-date ocean anomaly map reconstruction

The input is one NOAA global nighttime sea-surface-temperature anomaly map, dated 16 August 2026, with a 1991-2020 climatological baseline. Its NetCDF contains one time plane, a 0.01 degree C scale factor and explicit water, land, missing and ice flags. This is a processed satellite product.

The task reconstructs fixed water-pixel targets from twelve neighboring pixels on the same map. Target centers lie on a 40 pixel lattice; none of those centers can appear among any other target's calibration pixels. Complete 20 degree geographic tiles are assigned by a fixed hash, with 23 eligible tiles reserved.

The selected six-coefficient local stencil achieves 0.03284 degree C tile-balanced MAE and 0.06714 degree C worst-tile MAE. Unweighted local averaging gives 0.05750 and 0.2450 degree C. Eight attempts examined spherical distances, Taylor curvature cancellation, directional fronts, diagonal geometry, curvature limiting, latitude correction, range preservation and observation-noise averaging.

These results establish useful numerical reconstruction references. They do not establish ocean dynamics, future heat stress or coral bleaching thresholds. The same-map calibration and exclusion of incomplete coastal/ice neighborhoods are essential parts of the task, and no population confidence is inferred from dependent pixels.

Experimental scope and evaluation

Retrospective reconstruction of one daily map, using twelve fixed neighboring pixels from the same date. Not a future forecast.

Calibration: Cardinal neighbors at 4 and 8 pixels plus four diagonal neighbors at 4 pixels. Target centers lie on a 40 pixel lattice and can never be calibration pixels for any scored center.

Validation unit: Complete 20 degree geographic tile for scoring; all tiles belong to one dependent global map, not independent temporal replicates.

Tile-balanced errors and individual tile results only; no population confidence from dependent pixels.

Target: SST anomaly at a withheld pixel (degree C). Errors use degree C. The selected reference is **flexible**. Selection was frozen before confirmation; diagnostic winners are not substituted afterward.

Predictor	Unit
hx	degree C
hy	degree C
ox	degree C
oy	degree C
dg	degree C
gx	degree C
gy	degree C
clat	dimensionless cosine latitude

Matched comparison

Model	Development MAE	Reserved MAE	Reserved worst-group MAE
flexible	0.0356912	0.0328365	0.0671359
local_mean	0.0521489	0.0574952	0.245
zero_anomaly	0.853588	0.798152	2.05919
spherical_distance	0.0501236	0.0503713	0.134759
fourth_order_smooth	0.0398408	0.0401033	0.136667
front_adaptive	0.0507908	0.0496442	0.0785472
rotational_ninepoint	0.0444504	0.0454642	0.14875
limited_curvature	0.0392258	0.0378708	0.101033
latitude_curvature	0.0371352	0.0345851	0.0798387
range_preserving	0.0436794	0.0441662	0.136667
broad_noise_average	0.0837839	0.0952444	0.4075

The primary error is mean group MAE. RMSE and signed bias are complementary; rows within a group do not establish independent replication. No accuracy percentage or industrial tolerance is invented.

Selected equation and coefficients

$$\hat{T} = b_0 + b_1h_x + b_2h_y + b_3o_x + b_4o_y + b_5d_g$$

The h terms are inner horizontal/vertical neighbor means, o terms are the corresponding outer means, and dg is the diagonal-neighbor mean. All temperatures are in degrees Celsius; b0 carries that unit and the other coefficients are dimensionless. Explicit same-map calibration is required.

Coefficient	Frozen value
b0	0.0029814451
b1	1.02165722
b2	0.673113253
b3	-0.201321671
b4	-0.0982148325
b5	-0.397620849

All comparator expressions, numerical guards and full-precision values remain in EQUATIONS.md, rules.json and run.py.

Applicability and limitations

- Fixed complete-water neighborhoods exclude coasts, ice and missing neighborhoods.
- No climate evolution, causal ocean mechanism or coral bleaching hazard is identified. This is not the NOAA coral bleaching heat-stress product.

Only dimensions, flags, coordinates and scale metadata were inspected before this fixed geometry/hash split. All resulting scores become exposed.

Reproduction

Run `python run.py` in this final package to verify hashes and reproduce every saved prediction. Use `python run.py --inputs new.csv --output predictions.csv --model reference` only with the declared inputs and units. Numerical prediction evaluation does not certify novelty or mechanism. Source anchors and reserved observations are separate from predictor inputs.

Source: https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta_clim19912020.php. See `source/units` audit and prior-art records in research history.